

What determines the elongation of the bar?

↳ Stress and strain

These factors are studied in continuum mechanics

- Kinematics (Measure of deformation)
- Balance (equilibrium) → helps find measures of stress
- Constitutive Laws (stress-strain relations)

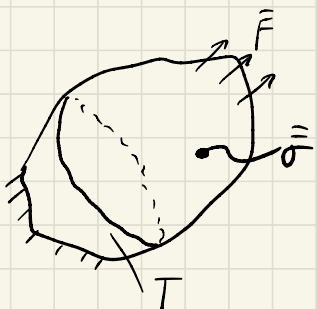


observed
not derived

— Math preliminaries

Tensors: A tensor is an entity which linearly transforms vectors

↳ physically: can represent physical quantities or properties that don't depend on a reference coordinate system.



↳ The ^{actual} physical stress at any point doesn't change when observed from two different frames, but the tensor does.

Scalars, vectors, tensors

Scalars: a $[2] \times 1$ matrix (just for completeness)

Vectors: V_i - indicial notation

$$[V] = [V_1, V_2, V_3, V_4, \dots]$$

$\bar{V}$ - direct notation

Tensor: σ_{ij} - indicial notation

$$[\sigma] = \begin{bmatrix} \sigma_{11} & \sigma_{12} & \sigma_{13} & \dots & \sigma_{1n} \\ \sigma_{21} & & & & \\ \vdots & & & & \\ \sigma_{n1} & & & & \sigma_{nn} \end{bmatrix}$$

$\bar{\bar{\sigma}}$ - direct notation (2nd order tensor)

Summation Convention

$$x = a_1 x_1 + a_2 x_2 + \dots + a_n x_n = \sum_{i=1}^N a_i x_i = \sum_{j=1}^N a_j x_j$$

Choice of letter for index is irrelevant (dummy index)

Using Summation Convention

$$x = a_i x_i = a_j x_j = \dots \text{ etc } \rightarrow \text{for dummy indices, summation is implied}$$

The value of N is either stated or known from context

Examples:

$$a_i x_i = a_1 x_1 + a_2 x_2 + a_3 x_3 \quad (n=3)$$

$$a_i \cdot a_i = a_1^2 + a_2^2 + a_3^2 + a_4^2 \quad (n=3)$$

$$\sigma_{ii} = \sigma_{11} + \sigma_{22} + \sigma_{33} \quad (n=3)$$

Notes: 1. A product having more than two occurrences of the same dummy index is meaningless (i.e. $a_i \cdot b_i \cdot c_i$)

2. A sum is always implied unless noted otherwise

3. In this class, $n=3$ so $i \in 1, 2, 3$ unless stated otherwise

An index that appears only once in each term is called a "free" index.

$$A_{ij} x_j = b_i = A_{i1} x_1 + A_{i2} x_2 + A_{i3} x_3 \quad \text{Thus,}$$

j is dummy
 i is free index

$$\begin{aligned} A_{11} x_1 + A_{12} x_2 + A_{13} x_3 &= b_1 \\ A_{21} x_1 + A_{22} x_2 + A_{23} x_3 &= b_2 \\ A_{31} x_1 + A_{32} x_2 + A_{33} x_3 &= b_3 \end{aligned}$$

or,

$$\begin{bmatrix} A_{11} & A_{12} & A_{13} \\ A_{21} & A_{22} & A_{23} \\ A_{31} & A_{32} & A_{33} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$$

Kronecker Delta δ_{ij}

Defined as $\delta_{ij} = \begin{cases} 1 & \text{if } i=j \\ 0 & \text{if } i \neq j \end{cases}$ Thus it is the identity matrix $\bar{I} = \delta_{ij}$

An important property of δ_{ij} is index substitution

↓

$$x_i \delta_{ij} = x_1 \delta_{1j} + x_2 \delta_{2j} + x_3 \delta_{3j}$$

$$= \begin{cases} x_1 & \text{if } j=1 \\ x_2 & \text{if } j=2 \\ x_3 & \text{if } j=3 \end{cases} \longrightarrow x_i \delta_{ij} = x_j$$

dummy is substituted with the free index

$$\text{EX: } A_{ij} \delta_{ij} = A_{11} = A_{11} + A_{22} + A_{33} = A_{jj}$$

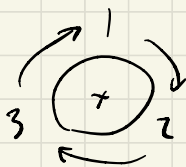
$$A_{ij} - A_{ik} \delta_{jk} = A_{ij} - A_{ij} = 0_{ij} \quad \left| \quad A_{ij} \delta_{jk} - A_{ij} \delta_{jk} = A_{ik} - A_{ik} = 0_{ik}\right.$$

↑ ↑
dummy k

$$\delta_{ii} = \delta_{11} + \delta_{22} + \delta_{33} = 3$$

Permutation symbol

defined as: $\epsilon_{ijk} = \begin{cases} 1 & \text{if } i, j, k \text{ form an even permutation of } 1, 2, 3 \\ -1 & \text{if } i, j, k \text{ form an odd permutation of } 1, 2, 3 \\ 0 & \text{if Do not form perm of } 1, 2, 3 \end{cases}$



$$\epsilon_{123} = \epsilon_{231} = \epsilon_{312} = 1$$

$$\epsilon_{132} = \epsilon_{321} = \epsilon_{213} = -1$$

$$\epsilon_{112} = \epsilon_{221} = \dots = 0$$

Operations with tensors

Addition $\bar{C} = \bar{A} + \bar{B} \longrightarrow C_{ij} = A_{ij} + B_{ij}$

Subtraction $\bar{C} = \bar{A} - \bar{B} \longrightarrow C_{ij} = A_{ij} - B_{ij}$

Magnification $\hat{\bar{B}} = \lambda \hat{\bar{A}} \rightarrow B_{ij} = \lambda A_{ij}$

Transpose $\hat{\bar{B}} = \hat{\bar{A}}^T \rightarrow B_{ij} = A_{ji}$

A tensor is symmetric if $\hat{\bar{B}} = \hat{\bar{B}}^T \rightarrow B_{ij} = B_{ji}$ (6 independent components from 9)

A tensor is skew if $\hat{\bar{B}} = -\hat{\bar{B}}^T \rightarrow B_{ij} = -B_{ji}$

must have 0 along the diagonal
(3 independent components)

We may write

$$B_{ij} = \underbrace{\frac{1}{2}(B_{ij} + B_{ji})}_{\text{Symmetric}} + \underbrace{\frac{1}{2}(B_{ij} - B_{ji})}_{\text{Skew}}$$

Every tensor admits a decomposition: $\hat{\bar{B}} = \text{sym } \hat{\bar{B}} + \text{skew } \hat{\bar{B}}$

$$\text{Sym } \hat{\bar{B}} = \frac{1}{2}(\hat{\bar{B}} + \hat{\bar{B}}^T) \quad \text{skew } \hat{\bar{B}} = \frac{1}{2}(\hat{\bar{B}} - \hat{\bar{B}}^T)$$

The tensor or Dyadic Product

$$\hat{\bar{A}} = \bar{x} \otimes \bar{y} \rightarrow A_{ij} = x_i y_j$$

Products:

Tensor by vector: $\bar{u} = \hat{\bar{A}} \cdot \bar{v} \rightarrow u_i = A_{ij} v_j \neq A_{ij} v_i$

Tensor on Tensor: $\bar{C} = \hat{\bar{A}} \cdot \hat{\bar{B}} \rightarrow C_{ij} = A_{ik} B_{kj}$ Thus, $\hat{\bar{A}} \hat{\bar{B}} \neq \hat{\bar{B}} \hat{\bar{A}}$

Inner product

$$\text{Vectors: } \bar{x} \cdot \bar{y} = \bar{y} \cdot \bar{x} = x_i y_i$$

$$\text{Tensor: } \bar{\bar{A}} : \bar{\bar{B}} = \bar{\bar{B}} : \bar{\bar{A}} = A_{ij} B_{ij}$$

Magnitude

$$\text{Vector: } \|\bar{u}\| = \sqrt{\bar{u} \cdot \bar{u}} = \sqrt{u_i u_i}$$

$$\text{Tensor: } \|\bar{\bar{A}}\| = \sqrt{\bar{\bar{A}} : \bar{\bar{A}}} = \sqrt{A_{ij} A_{ij}}$$

Cross product:

$$\text{Vectors: } \bar{a} = \bar{u} \times \bar{v} = a_i \epsilon_{ijk} u_j v_k$$

$$a_i = \epsilon_{123} u_2 v_3 + \epsilon_{132} u_3 v_2 + \epsilon_{111} u_1 v_1 + \epsilon_{112} u_1 v_2 \\ + \epsilon_{113} u_1 v_3 + \dots \quad \text{repeating indices are zero}$$

$$a_i = u_2 v_3 - u_3 v_2$$

Trace of a tensor

$$\text{Tr}(\bar{A}) = A_{ii} \quad (\text{sum of components on diagonal})$$

$$\text{Thus, } \text{tr}(\bar{x} \otimes \bar{y}) = \bar{x} \cdot \bar{y} \rightarrow \text{tr}(x_i y_j) = x_i y_i$$

A tensor $\bar{\bar{A}}$ is deviatoric if: $\text{Tr}(\bar{\bar{A}}) = 0$

We define the deviatoric part of a tensor $\bar{\bar{A}}$ as:

$$\bar{\bar{A}}' = \text{Dev} \bar{\bar{A}} = \bar{\bar{A}} - \frac{1}{3} \text{tr}(\bar{\bar{A}}) \bar{\bar{I}}$$

or

$$A'_{ij} = A_{ij} - \frac{1}{3} A_{kk} \delta_{ij}$$

$$\begin{aligned} \text{Tr}(\bar{\bar{A}}') &= \text{tr}(A_{ij} - \frac{1}{3} A_{kk} \delta_{ij}) = A_{pp} - \frac{1}{3} A_{kk} \delta_{pp} \\ &= A_{pp} - \frac{1}{3} A_{kk} (3) = A_{pp} - A_{kk} = 0 \end{aligned}$$

In doing this, we defined the spherical part of $\bar{\bar{A}}$

$$\frac{1}{3} \text{tr}(\bar{\bar{A}}) \bar{\bar{I}} \leftarrow \text{spherical part}$$

Every tensor then admits a decomposition in to deviatoric and spherical parts...

$$\begin{array}{ccc} \bar{\bar{A}} & = & \bar{\bar{A}}' + \frac{1}{3} \text{tr}(\bar{\bar{A}}) \bar{\bar{I}} \\ \uparrow & & \uparrow \\ \text{dev} & & \text{spherical} \end{array}$$

Aside: FREE Indices have to match on each side of equation.

Determinant of a tensor

$$\det(\bar{\bar{S}}) = \frac{\bar{\bar{S}} \bar{\bar{u}} \cdot (\bar{\bar{S}} \bar{\bar{v}} \times \bar{\bar{S}} \bar{\bar{w}})}{\bar{\bar{u}} \cdot (\bar{\bar{v}} \times \bar{\bar{w}})} \quad \text{For any Non-coplanar } \{\bar{\bar{u}}, \bar{\bar{v}}, \bar{\bar{w}}\}$$

Invertible tensor

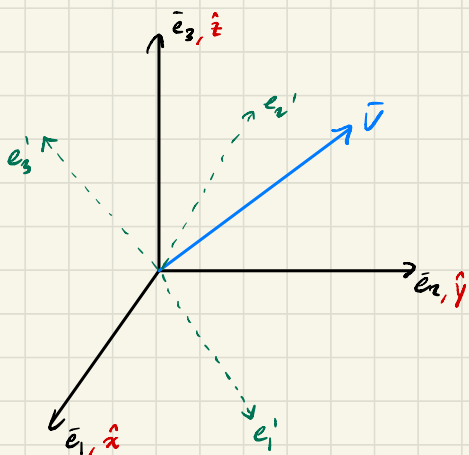
$\bar{\bar{S}}$ is invertible if there exists an inverse $\bar{\bar{S}}^{-1}$ such that:

$$\bar{\bar{S}} \bar{\bar{S}}^{-1} = \bar{\bar{S}}^{-1} \bar{\bar{S}} = \bar{\bar{I}} \rightarrow \bar{\bar{S}} \bar{\bar{B}} = \bar{\bar{B}} \bar{\bar{S}} = \bar{\bar{I}} \text{ then } \bar{\bar{B}} = \bar{\bar{S}}^{-1}$$

$\bar{\bar{S}}$ is invertible if and only if $\det(\bar{\bar{S}}) \neq 0$

Components and change of Basis

- Vectors and tensors are physical entities independent of the frame we choose to represent them
- If we know the components with respect to one frame, we know them with respect to another



$$\text{Basis: } \bar{e}_1 \cdot \bar{e}_2 = 0 \quad \bar{e}_1 \cdot \bar{e}_1 = 1$$

$$\bar{e}_1 \cdot (\bar{e}_2 \times \bar{e}_3) = 1$$

$$\bar{e}_i \cdot \bar{e}_j = \delta_{ij}$$

i, j not index, it is a label of which unit vector we are using

$$\bar{e}_i \cdot (\bar{e}_j \times \bar{e}_k) = \epsilon_{ijk}$$

$$\bar{V} = V_1 \bar{e}_1 + V_2 \bar{e}_2 + V_3 \bar{e}_3 = V_i \bar{e}_i$$

$$\bar{V} = V_1' \bar{e}_1' + V_2' \bar{e}_2' + V_3' \bar{e}_3' = V_i' \bar{e}_i'$$

Find V_i' in terms of V_i

$$\bar{V} \cdot \bar{e}_i' = V_i' \quad \text{as } V_i' = \bar{V} \cdot \bar{e}_i' = (V_j \bar{e}_j) \cdot \bar{e}_i' \\ = (\bar{e}_i' \cdot \bar{e}_j) V_j \\ \boxed{V_i' = Q_{ij} \cdot V_j}$$

Q_{ij} is a Rotation tensor

$$\bar{U}' = \bar{Q} \bar{U}, \text{ where } Q_{ij} = \bar{e}_i' \cdot \bar{e}_j$$

Similarly...

$$\begin{aligned} V_i &= \bar{V} \cdot \bar{e}_i = (V_j' \bar{e}_j') \bar{e}_i = (\bar{e}_i \cdot \bar{e}_j') V_j' \\ &= (\bar{e}_j' \cdot \bar{e}_i) V_j' \\ &= Q_{ji} V_j' \end{aligned}$$

$$\bar{V} = \bar{Q}^T \bar{U}'$$

All rotation Tensors obey:

$$\bar{Q} \bar{Q}^T = \bar{Q}^T \bar{Q} = \bar{I} \quad \text{and} \quad \det(\bar{Q}) = \pm 1$$